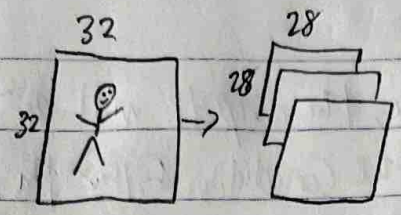
#4 Jan 21

a)



b) $5 \times 5 \times 3 = 75$ parameters, 25 in each filter w/ 3 separate filters

c) we can think of this as an ordinary feed forward network by having the 32x32 pixels stretched out to form the input vector, we would then carefully assign each of the input nodes to the 75 nodes in the hidden "convolution" layer in such a way that the first 25 of the convolution layer correspond to the different 5x5 sections of the input "image". The first 25 weights are the first filter, the second 25 are 2nd, & 3rd 3rd. There would be minimal 0 weights, only in the "corners" of the input "image". There will only be 75 total unique weights, assuming all the weights in the 3 conv. filters are unique.

d) w/ no constraints, $32 \times 32 = (1024 + \overset{\text{bias}}{1}) \times 75 = 76{,}875$ weights! wow, over a factor of 100!

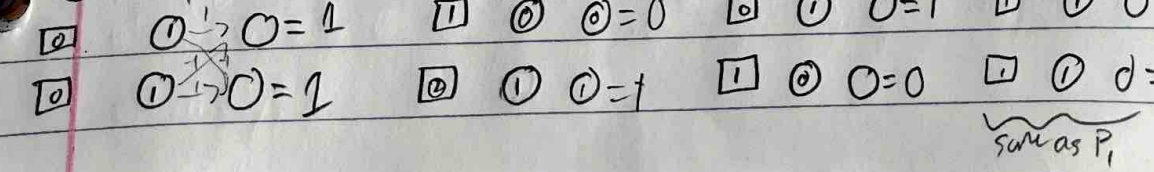
#8-12 #8 Model A has less parameters because there are less total nodes, meaning it will likely train faster. Model B, even though it has an extra layer, could be more understandable since the weights of the liner layer give us data about how each of the inputs are reacting w/ & affecting the output.

#9 8 input patterns, like the 3 values on a tuba correspond to the 7 positions on a trombone. 4 inputs gives us 16 different combs, 5 gives us 32. The formula is 2^n, where n is the # of inputs

plus one value

#10 $P_1 = g(2 \cdot 1) = 1$, $P_2 = g(1 \cdot -4 + 1 \cdot 1) = 0$, $P_3 = g(2 \cdot 1 + 1 \cdot 1) = 1$, $P_4 = g(2 - 4 + 1) = 0$

#11 P_1: $g(1-1)=1$ P_2: P_3: P_4



#12 At the very top & bottom of the circle, the outputs will be -4 & 4 respectively, since the white pixels affect the Convolution filter the most, and there's no subtraction due to differing signs (like at the rest of the middle crossing of the circle. The x location will take on a negative value, since its on the edge in the upper half of the circle, meaning none of the top layer of the filter will be active with the white 1s that the bottom.